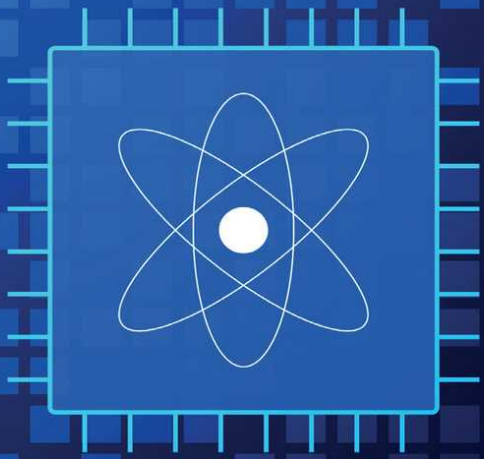

QUANTUM COMPUTING STRATEGY

FOUNDATIONS AND APPLICABILITY

Elena Yndurain

A **Chapman & Hall** Book

CRC Press
Taylor & Francis Group

Quantum Computing Strategy

Quantum computing is not merely an incremental advancement in computing technology; it represents a fundamentally new paradigm, distinct from classical computing. Rooted in quantum mechanics, it introduces an entirely novel information theory. As a result, translating existing models, solution designs, and approaches to quantum computing is a complex, non-trivial task. This comprehensive book demystifies quantum concepts through accessible explanations, practical case studies, and real-world examples from industries such as aerospace, agriculture, automotive, chemicals, energy, finance, government, healthcare, manufacturing, supply chain, and telecommunications.

Blending a business perspective with a scientific rigor, this book is divided into two parts. The first part covers foundational technical concepts, including quantum mechanics principles that enable quantum technologies, key quantum algorithms, mathematical frameworks, quantum computing technologies, post-quantum cryptography, the types of problems quantum computers solve, and the technology's outlook. The second part focuses on practical applicability, presenting industry use cases, guidance on approaching quantum computing problems, mapping use cases to quantum computing, responsible quantum computing practices, and a roadmap for businesses preparing for quantum adoption. This structured approach equips readers with the knowledge and tools to effectively integrate quantum computing into their strategic planning.

Quantum Computing Strategy: Foundations and Applicability serves as an essential reference for technology enthusiasts, business leaders, policymakers, and educators seeking to understand the benefits quantum computing offers enterprises. Designed as a self-contained learning resource, it empowers readers to navigate the emerging quantum landscape confidently.

Elena Yndurain is a high-tech executive and professor with expertise in operationalizing innovation. She holds a Ph.D. and M.Sc. in Telematics Engineering focused on Artificial Intelligence, as well as an Executive MBA and a B.Sc. in Computer Science and Mathematics. With extensive international experience in consulting, technology, and banking, she has successfully launched numerous technology products. She was a pioneer in teaching quantum computing to business executives. Most recently, she served as the European CEO of a quantum computing software startup and is an active member of international advisory boards focused on technology governance.

Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

Quantum Computing Strategy

Foundations and Applicability

Elena Yndurain

CRC Press
Taylor & Francis Group
Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business
A CHAPMAN & HALL BOOK

Illustrations: Jorge Zeballos.

First edition published 2025
by CRC Press
2385 Executive Center Drive, Suite 320, Boca Raton, FL 33431

and by CRC Press
4 Park Square, Milton Park, Abingdon, Oxon, OX14 4RN

CRC Press is an imprint of Taylor & Francis Group, LLC

© 2025 Elena Yndurain

Reasonable efforts have been made to publish reliable data and information, but the author and publisher cannot assume responsibility for the validity of all materials or the consequences of their use. The authors and publishers have attempted to trace the copyright holders of all material reproduced in this publication and apologize to copyright holders if permission to publish in this form has not been obtained. If any copyright material has not been acknowledged please write and let us know so we may rectify in any future reprint.

Except as permitted under U.S. Copyright Law, no part of this book may be reprinted, reproduced, transmitted, or utilized in any form by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying, microfilming, and recording, or in any information storage or retrieval system, without written permission from the publishers.

For permission to photocopy or use material electronically from this work, access www.copyright.com or contact the Copyright Clearance Center, Inc. (CCC), 222 Rosewood Drive, Danvers, MA 01923, 978-750-8400. For works that are not available on CCC please contact mpkbookspermissions@tandf.co.uk

Trademark notice: Product or corporate names may be trademarks or registered trademarks and are used only for identification and explanation without intent to infringe.

ISBN: 9781032299143 (hbk)
ISBN: 9781032275826 (pbk)
ISBN: 9781003302674 (ebk)

DOI: 10.1201/9781003302674

Typeset in Minion
by codeMantra

To my father—you would have enjoyed this book

Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

Contents

Preface, xiii

Acknowledgments, xviii

Glossary, xx

Acronyms, xxviii

INTRODUCTION 1

SECTION I **Foundations** **5**

CHAPTER 1 ▪ Key Concepts 9

1.1 QUANTUM INFORMATION 10

1.2 QUANTUM PROPERTIES 12

1.3 MATHEMATICAL DEFINITIONS 16

CHAPTER 2 ▪ Quantum Computers Overview 18

2.1 ANALOG AND DIGITAL QUANTUM COMPUTERS 20

2.2 QUANTUM COMPUTER SIMULATORS 22

2.3 QUBIT MODALITIES DEFINITIONS 23

2.4 DIFFERENT QUBIT TOPOLOGY 28

CHAPTER 3 ▪ Quantum Programming 30

3.1 CIRCUITS DEFINITION 31

3.2 GATES FUNCTIONING 32

3.3 CIRCUIT REQUIREMENTS 35

3.4 PROGRAMMING TOOLS 37

CHAPTER 4 ▪ Quantum Algorithms Overview 39

4.1 HYBRID APPROACH 40
4.2 SOFTWARE APPROACH 41
4.3 HARDWARE APPROACH 41
4.4 QUANTUM INSPIRED ALGORITHMS 42
4.5 ALGORITHM PERFORMANCE 43
4.6 ALGORITHM NOISE 46

CHAPTER 5 ▪ Algorithm Foundations 50

5.1 GROVER UNSTRUCTURED SEARCH 52
5.2 SHOR NUMBER FACTORIZATION 53
5.3 DEUTSCH–JOZSA ORACLE 54
5.4 BERNSTEIN–VAZIRANI DECODE 55
5.5 HARROW–HASSIDIM–LLOYD LINEAR SOLVERS 55
5.6 QUANTUM METROPOLIS EQUILIBRIUM 56

CHAPTER 6 ▪ Applied Algorithms 57

6.1 VARIATIONAL QUANTUM EIGENSOLVER (VQE) 58
6.2 QUANTUM AMPLITUDE ESTIMATION (QAE) 58
6.3 QUANTUM APPROXIMATE OPTIMIZATION ALGORITHM (QAOA) 59
6.4 QUADRATIC UNCONSTRAINED BINARY OPTIMIZATION (QUBO) 60
6.5 QUANTUM DIFFERENTIAL EQUATION (QDE) 60
6.6 QUANTUM ARTIFICIAL INTELLIGENCE (QAI) 61
6.7 ALGORITHMS INTERRELATION 62

CHAPTER 7 ▪ Problem Categorization 64

7.1 CHEMICAL PROCESSES 66
7.1.1 Classical Approach 66
7.1.2 Business Impact 66
7.1.3 Quantum Approach 67

7.1.4 Specific Example 67
7.1.5 Simplified Analogy 68
7.2 OBJECTIVE OPTIMIZATION 69
7.2.1 Classical Approach 69
7.2.2 Business Impact 70
7.2.3 Quantum Approach 70
7.2.4 Specific Example 71
7.2.5 Simplified Analogy 71
7.3 STOCHASTIC SIMULATIONS 72
7.3.1 Classical Approach 73
7.3.2 Business Impact 73
7.3.3 Quantum Approach 73
7.3.4 Specific Example 74
7.3.5 Simplified Analogy 74
7.4 ARTIFICIAL INTELLIGENCE 75
7.4.1 Classical Approach 76
7.4.2 Business Impact 76
7.4.3 Quantum Approach 76
7.4.4 Specific Example 77
7.4.5 Simplified Analogy 78

CHAPTER 8 ■ Quantum Computing Risk 79

8.1 QUANTUM CRYPTOGRAPHIC SCHEMES 81
8.2 QUANTUM-SECURE CRYPTOGRAPHY QKD 82
8.3 POST-QUANTUM CRYPTOGRAPHY ALGORITHMS 84
8.4 QUANTUM SAFETY STRATEGY PLAN 86
8.5 SECURITY USE CASES 87

CHAPTER 9 ▪ Technology Adoption Outlook 89

SECTION II **Applicability** **95**
ANALYSIS METHODOLOGY 98

CHAPTER 10 ▪ Aerospace 99
10.1 USE CASE: ENGINE DESIGN 101
10.2 USE CASE: AIR CARGO LOAD 102
10.3 USE CASE: IRREGULAR OPERATIONS 103
10.4 USE CASE: FLIGHT CLIMB 104

CHAPTER 11 ▪ Agriculture 105
11.1 USE CASE: EFFICIENT FERTILIZERS 107
11.2 USE CASE: SEED BIOLOGY 108
11.3 USE CASE: WEATHER FORECAST 109
11.4 USE CASE: IMPROVED CROP YIELD 110

CHAPTER 12 ▪ Automotive 111
12.1 USE CASE: EV BATTERIES/FUEL CELLS 113
12.2 USE CASE: TRANSPORT ROUTING FLOW 114
12.3 USE CASE: OBJECT DETECTION 115
12.4 USE CASE: AERODYNAMIC DESIGN 116

CHAPTER 13 ▪ Chemicals 117
13.1 USE CASE: UNDERSTANDING MOLECULAR PROPERTIES 119
13.2 USE CASE: DESIGN OF AGGREGATES 120
13.3 USE CASE: CRYSTAL STRUCTURE 121
13.4 USE CASE: CHEMICAL REACTIONS CATALYSTS 122

CHAPTER 14 ▪ Energy 123
14.1 USE CASE: RESERVOIR SIMULATION 125
14.2 USE CASE: ENERGY UNIT COMMITMENT 126

14.3 USE CASE: SMART-GRID OPERATION 127
14.4 USE CASE: GAS TURBINE DESIGN 128

CHAPTER 15 ■ Finance 129
15.1 USE CASE: PORTFOLIO MANAGEMENT 131
15.2 USE CASE: FRAUDULENT TRANSACTIONS 132
15.3 USE CASE: PRODUCT PRICING ACCURACY 133

CHAPTER 16 ■ Government 134
16.1 USE CASE: CARBON CAPTURE SUSTAINABILITY 136
16.2 USE CASE: TRANSPORT EFFICIENCY 137
16.3 USE CASE: SATELLITE IMAGING 138
16.4 USE CASE: MILITARY OPERATIONS 139

CHAPTER 17 ■ Healthcare Life Sciences 140
17.1 USE CASE: DRUG CANDIDATES 142
17.2 USE CASE: MEDICAL IMAGING 143
17.3 USE CASE: PROTEIN PATHOLOGY 144
17.4 USE CASE: DRUG TOXICITY 145

CHAPTER 18 ■ Manufacturing 146
18.1 USE CASE: IMPROVING MATERIALS 148
18.2 USE CASE: ASSEMBLY LINE FLOW 149
18.3 USE CASE: PREDICTIVE MAINTENANCE 150
18.4 USE CASE: COMPONENTS PERFORMANCE 151

CHAPTER 19 ■ Supply Chain 152
19.1 USE CASE: ENERGY DELIVERY 154
19.2 USE CASE: LOAD OPTIMIZATION 155
19.3 USE CASE: JUST-IN-TIME LOGISTICS 156
19.4 USE CASE: DEMAND FORECAST 157

CHAPTER 20 ■ Telecommunications 158

20.1 USE CASE: RADIO ACCESS 160
20.2 USE CASE: NETWORK PLANNING 161
20.3 USE CASE: SERVICE QUALITY 162
20.4 USE CASE: MIMO SPECTRUM EFFICIENCY 163

CHAPTER 21 ■ Use Case Problem Mapping 164

21.1 EXAMPLE PROBLEM: PACKAGE DELIVERY 164

CHAPTER 22 ■ How to Get Quantum Ready 168

22.1 STRATEGY: WHERE AM I NOW AND HOW FAR DO I WANT TO GO? 169
22.2 TEAM: DO PEOPLE HAVE THE TIME AND SKILLS TO WORK ON THIS? 170
22.3 USES: HOW TO DEFINE, ANALYZE, PRIORITIZE, AND SELECT USE CASES? 170
22.4 PROVIDER: HOW TO SELECT THE BEST PROVIDER FOR ME? 171
22.5 TESTS: WHICH TECHNOLOGY WILL I NEED TO SELECT TO EVALUATE THE USE CASES? 173

CHAPTER 23 ■ Responsible Technology Use 174

23.1 DISTRIBUTE BENEFITS EQUITABLY 175
23.2 REDUCE ENVIRONMENTAL IMPACT 175
23.3 ENSURE ALGORITHM ACCOUNTABILITY 176
23.4 RESKILL EMPLOYEES 176
23.5 PREVENT HOSTILE USE 176
23.6 SECURITY DILEMMA 176

RESOURCES, 179

BIBLIOGRAPHY, 181

INDEX, 199

Preface

This book, *Quantum Computing Strategy: Foundations and Applicability*, provides an overview of quantum technologies, focused on quantum computing, including specific use case applicability examples and concept definitions with the help of some simplification visualizations to facilitate the understanding from a high-level perspective.

This book marks not only the culmination of years of study and exploration but also a deeply personal journey into the world of quantum computing. This journey, enriched by a set of important experiences, has shaped not only my understanding of quantum computing but also my perspective on its strategic implications for business applicability.

MY JOURNEY INTO QUANTUM COMPUTING

My first encounter with quantum computing was in October 2000, when my father invited me to a talk by Spanish physicist Ignacio Cirac, who had defined the first trapped-ion quantum computer. In his talk, "Is it possible to build a quantum computer?", he explained the concept of quantum information, what a quantum computer is, and how we can build it. I was intrigued by this technology, which was so different from what had come before. In this conference, I was surprised to run into Jaime Botín, founder of Bankinter, a major Spanish commercial bank; he most likely had an intuition about how important quantum computing would be for the financial sector.

It was not until May 2016 that quantum computing came back into my life. At that time, IBM opened its quantum computer to the world via the cloud, providing access to early quantum computing prototypes to foster the creation of use cases and the development of future applications, as publicly stated by the company at that time.

I saw real quantum computers in a visit to IBM's quantum computer at their Thomas J. Watson Research Center in Yorktown Heights, NY,

and to the Joint Quantum Institute (JQI) at the University of Maryland. I had the chance to get a first-hand overview of the computer in a lab tour by the researchers who were building the computer at both institutions – Jay Gambetta, Jerry Chow, and Mark Ritter from IBM as well as Prof. Luis Orozco and Prof. Mohammed Hafezi from JQI. At the time, none of these researchers were famous yet in the industry, and their language was candid and physicist-like, quite different from what there is today, which is an example of how quantum computing has gone to the forefront of business.

After these visits, one thing was clear to me: this technology was very different from other technologies. Quantum technologies are heavily based on quantum physics, and I decided to learn it so I could understand it better.

In 2017, I contacted Miguel Ángel Martín-Delgado, a full professor of theoretical physics at Complutense University of Madrid (UCM) and an expert in quantum information and error correction codes. He told me he believed that the next decade would be known as "the golden quantum twenties" and that expectations would be set for the so-called "second quantum revolution". In a series of meetings, he very generously taught me the foundations of quantum information and ignited my interest in self-study.

That same year, I added quantum computing to my own teaching repertoire. I created and taught a business overview course at IE Business School, targeted at master students from different programs. This was a first-of-its-kind initiative. It was one of the first courses where quantum computing was distilled into a format accessible to business students, focusing on its strategic implications rather than just its technical aspects. This course aimed to demystify quantum computing for future business leaders, enabling them to grasp how this technology could drive innovation, create new markets, and influence competitive industry dynamics.

In 2018, I got fully immersed in quantum computing by joining the early IBM Quantum team to work on business applicability and problem categorization. This work led me to become a quantum computing expert for startups, industry corporations, and academia, helping them become quantum ready at different levels.

A couple of years later, I created a more technical course for the Universidad Carlos III de Madrid (UC3M) Telecommunications Engineering and Cybersecurity program. The course, while more technical, was also designed with a focus on practical applications. This was a

pioneering approach in that it provided engineering students with not just the theoretical foundations of quantum computing but also an understanding of how these principles could be applied to solve complex problems in their fields, such as secure telecommunications and data protection.

Until today, I continue to evangelize quantum computing through different universities, market research institutions, and corporations such as Stevens Institute of Technology, The University of Strasbourg, Arcano Economic Research, UC3M, Indra Sistemas, PG&E Corporation, and IE University, just to name some.

In my journey as both a learner and educator in the field of quantum computing, I have encountered a wide array of questions from students and executives alike. These recurring inquiries highlighted a common challenge: the need for a more accessible explanation of quantum computing concepts that would resonate with those outside the field of physics. Recognizing this, I embarked on a mission to demystify the complexities of quantum computing for a non-physicist audience.

The task at hand was not trivial. Quantum computing, with its roots deeply embedded in the principles of quantum mechanics, is inherently abstract and counterintuitive. The fundamental challenge lay in translating these intricate theories into concepts that are intuitive and relatable. To address this, I started by identifying the core elements of quantum computing that most captivated the curiosity of newcomers, regardless of their background or level of expertise. These ranged from the basic principles, like superposition and entanglement, to more advanced topics such as quantum algorithms and their potential applications.

One of the most common requests from learners, both in academic settings and the industry, was for a simplified visualization of quantum mechanics' foundational concepts. They sought to grasp not only how these principles differentiate quantum computing from classical computing but also the unique advantages and potential that quantum computing holds. To meet this need, I employed a variety of educational tools and analogies that resonate with everyday experiences, making the abstract concepts more tangible. For instance, I likened the principle of superposition to the multiple potential states of a spinning coin before it lands, offering a glimpse into the quantum world's probabilistic nature.

Moreover, there was a clear demand for real-world examples that illustrate the practical applications of quantum computing. Learners wanted to see beyond the theoretical framework to understand how this groundbreaking technology could address complex challenges across various

industries. In response, I curated a collection of case studies and potential use cases, from drug discovery and material science to optimization problems and cryptography. These examples not only showcased the broad applicability of quantum computing but also highlighted its potential to revolutionize industries by solving problems that are currently intractable for classical computers.

STRUCTURE OF THIS BOOK

This book is split into two sections. The first section lays out the key concepts of quantum computing to ensure we understand this fascinating yet counterintuitive technology. It defines what makes quantum computing so special to set the seed for the *foundation* for the rest of this book. It covers the definition of algorithms explaining their key advantages and the underlying concepts in a high-level business-oriented way so that anyone can understand them. The second section covers the different quantum computing solutions providing *applicability* examples for different industries. It explains the impact of this technology from a business angle and how to get quantum ready from a corporation point of view.

HOW TO USE THIS BOOK

This book is designed to be entirely self-contained, meaning that it does not rely on external resources, additional texts, or prior extensive knowledge from the reader to provide a comprehensive understanding of quantum computing.

This approach ensures that readers can access and understand the full scope of quantum computing and its applications independently. The text includes a variety of real-world examples that demonstrate how quantum computing is applied in different industries and scientific research. These examples help to bridge the gap between abstract theoretical concepts and practical applications, making the material more accessible and engaging.

Additionally, this book incorporates numerous visualizations and illustrative diagrams. These visual aids are specifically designed to help clarify complex concepts and provide intuitive insights into the mechanics of quantum systems and algorithms. By using these visual tools, this book aids readers in grasping sophisticated ideas more effectively, enhancing their overall learning experience.

Overall, this book is meticulously organized to guide readers through the special world of quantum computing, from the basic principles to

the more advanced topics, without the need for additional support. It is designed to bridge the gap between abstract quantum theory and practical computational strategies. I hope that this work will help students and professionals who are interested in new technologies in their learning process.

Acknowledgments

There are many people who have helped me with this book, directly or indirectly, through specific discussions about this book's topics, discussions of quantum computing in general, or simply with moral encouragement in difficult times.

Many generous people helped me peer review the content of my manuscript, brainstorming on industry analysis, use case challenges, quantum computing impact, or underlying quantum information concepts. In particular, the following people took time from their busy schedules to discuss or review some of the topics I was analyzing in the book on industry business challenges or quantum technology concepts.

They are listed here alphabetically by first name: Alejandro Peña Galiano, aerospace engineer; Alistair Nolan, OECD; Andrea Rodríguez-Blanco, post-doc at the University of California Berkeley; Antonio J. Montero Sánchez, telecommunications expert at MASMOVIL Group (now MASORANGE); Bob Sutor, ex-IBM research executive; Cristina Sánchez Muñoz, expert in automotive industry, manufacturing, and digitalization; Daniel Egger, quantum computing expert; David Muñoz, quantum chemistry expert; Emma Wang, expert in energy, OECD; Escolástico Sánchez, head of Quantum Computing, BBVA; Ewan Munro, expert in quantum information, COO Entropica Labs; Francisco Martín-Fernández, STSM at IBM Quantum; Hector Guerrero, expert in government and defense; Javier Echenique, expert in supply chain, CEO at IDLogistics Italy; José Ramón Tora, expert in financial services; Kai Watanabe, expert in quantum computing technology and business strategy; Kesha Sorathia, computational chemistry expert; Kumar Bhaskaran, IBM researcher; Laura Barea, expert in pharma at AbbVie; Lauri Toikka, quantum information researcher; Marwa Farag, quantum computing and quantum chemistry expert; Michael D. Cohen, technology strategy expert; Michele Grossi, coordinator quantum algorithm at CERN,

Miguel Ángel Martín-Delgado, quantum information expert, Universidad Complutense de Madrid; Ricardo Enriquez, Quantum Technologies for industry expert; Ricardo Olalla, expert in agriculture and digitalization, BOSCH; Rima Kasia Oueid, expert in government and quantum commercialization; Roman Orús Ikerbasque Quantum Information Professor and Multiverse Computing co-founder, Sharana Kariappa, optics engineer at Nu Quantum; Tom Lubinski, quantum programming software architect at Quantum Circuits Inc; and Veronica Fernandez-Mármol, quantum communications expert and CSIC tenured scientist.

I would like to give Tom Lubinski special thanks for spending additional time brainstorming and reviewing some key concepts of this book.

I also want to mention the assistance from IE Business School students who researched some of the information that I needed as a source of analysis: Rafael Cifuentes, Ashley Duffins, Ishita Kishore, Ramez Kabbani, Alex VanOeveren, and Marnix Van der Stighelen, with special mention to fellow student Abdullah Ruwaili from McKinsey & Company, who collaborated on the use case business impact analysis.

Apart from the people who have directly contributed to this book, there have been others who have indirectly done so, from whom I learned about different quantum technology topics in the different industry events that I have attended or watched online, such as the IEEE Quantum Week led by Hausi Muller, the Quantum Economic Development Consortium (QED-C) led by Celia Merzbacher, the Q2B conference led by Matt Johnson, and the Commercialising Quantum Global conference led by the Economist. In some of those events, apart from participating, I was also able to discuss quantum technologies with other experts. I had the opportunity to hear about the state-of-the-art of quantum technology from quantum computing companies, investors, industrial corporations, academics, researchers, and government representatives.

Of course, I want to extend my thanks to my publisher, Randi Slack, for her patience and support while I prepared the book's manuscript and to the Taylor & Francis Group for its interest in helping advance Quantum Computing adoption making it accessible to all audiences.

Finally, I again want to thank Prof. Miguel Ángel Martín-Delgado for suggesting and encouraging me to write this book to help link research to business.

Glossary

A

Aerodynamics: A branch of fluid dynamics that studies how gases interact with moving bodies to understand how objects move through the air.

Advanced Encryption Standard: A symmetric encryption algorithm widely used across the globe.

Air Cargo Load: Weighted, correctly loaded, and properly secured cargo in aircraft to prevent movement in flight.

Air Traffic: The aircraft, or the number of aircraft, flying in an area or along a route

Annealer: A special type of quantum computer designed to solve optimization problems.

Autonomous Vehicles: Self-driving cars capable of sensing their environment using sensors and moving safely with little or no human input.

B

Bell State: A specific type of quantum state that involves two qubits in a maximally entangled state, named after the physicist John Stewart Bell.

Binding Affinity: Strength with which a drug binds to its target involved in a disease process in the body, usually a protein such as an enzyme or receptor.

Bloch Sphere: A geometrical representation of the pure state space of a two-level quantum mechanical system (qubit), named after the physicist Felix Bloch.

Business Advantage: Anything that gives a company an edge over its competitors, in terms of cost advantage, attracts more customers, or grows its market share.

C

Carbon Capture Utilization: Techniques to use a harmful waste product (CO_2, which contributes to climate change) in industrial processes by capturing carbon dioxide emissions from generation sources.

Central Processing Unit: It is the most important processor in a computer, it has electronic circuitry that executes instructions of a computer program.

Catalyst: A substance that increases the rate of a chemical reaction without itself undergoing any permanent chemical change.

Coherence: Ability of a quantum state to maintain its entanglement and superposition in the face of interactions and the effects of thermalization.

Cryptographic Agility: Ability of security hardware to update an algorithm without the need to rewrite applications or deploy new hardware systems.

CRYSTALS-Kyber: A lattice-based cryptographic algorithm.

D

Decoherence: The process by which quantum properties decay over time.

Derivatives: Financial contracts whose value is dependent on an underlying asset, group of assets, or benchmark.

Drug Discovery: The process of identifying a new medication by finding compounds that can effectively treat or prevent diseases.

Dynamic Routing: A networking technique used to find the best path for data to travel across a network that automatically adjusts the paths.

E

Electric Battery: A source of power that converts chemical energy into electrical energy through an electrochemical reaction and is rechargeable.

Elliptic Curve Cryptography: Form of public-key cryptography based on the algebraic structure of elliptic curves over finite fields.
European Telecommunications Standards Institute (ETSI): Independent, not-for-profit, standardization organization in the field of information and communications.

F

Fault-Tolerant Quantum Computer (FTQC): An error-corrected quantum computing system designed to perform accurate computations.
Flight Climb: The operation of increasing the altitude of an aircraft.
Fraud: Deceptiveness or misrepresentations to gain a financial or personal advantage, or to cause harm to others.

G

Genomic Analysis: Study of DNA sequencing, structural variation, gene expression, or regulatory and functional element annotation at a genomic scale.
Gross domestic product (GDP): A monetary measure of the market value of all the final goods and services produced and rendered in a time period by a country.

H

Hamiltonian: A fundamental concept of quantum mechanics that represents the total energy of a system kinetic and potential.
High-Performance Computing (HPC): Use of supercomputers and computer clusters to solve advanced computation problems.
Hilbert Space: A mathematical framework for representing and manipulating quantum states, named after the mathematician David Hilbert.

I

Information and Communication Technology (ICT): Infrastructure and components that enable computing to manage information and facilitate communication.
Irregular Operations (IROPs): Disruptions causing customers not to be able to use the flight(s) ticketed.

International Organization for Standardization (ISO): Independent, non-governmental, international standard development organization composed of representatives from the national standards organizations of member countries, HQ in Europe.

K

Key Encapsulation Mechanism: A cryptographic protocol used in securing communication between two parties by encapsulating a secret key.

KET: A column vector in the space of quantum states to describe the state of a quantum system.

L

Light weighting Design: Constructing cars and trucks with lighter designs to enhance fuel efficiency and improve handling.

Linear Algebra: A branch of mathematics about linear equations fundamental to quantum computing to describe qubit states and perform quantum operations.

Logical Qubit: A combination of qubits that work together to perform computations with a lower error rate.

M

Machine Learning: A branch of artificial intelligence that focuses on using data and algorithms that are able to learn and adapt without following explicit instructions.

Material Compounds: Compounds made from two or more constituent materials with different physical or chemical properties.

Mean time between failure (MTBF): Measure of the reliability of a system or component used in the context of industrial or electronic system maintainability.

Molecular Energy States: All the energy forms that molecules can possess due to their atomic arrangements and movements, which exist at a specific energy level.

Multiple Input Multiple Output: A wireless communication technology that uses multiple antennas to increase channel capacity.

N

National Institute of Standards and Technology (NIST): An agency of the US Department of Commerce with the mission to promote national innovation and industrial competitiveness.

National Security Agency/Central Security Service (NSA/CSS): Leads the U.S. Government in cryptology encompassing both signals intelligence and cybersecurity.

Noisy Intermediate-Scale Quantum (NISQ): A class of quantum computing devices that are currently in development and used with significant errors in them.

P

Post-quantum Cryptography: Cryptographic algorithms that are believed to be secure against an attack by a quantum computer, i.e., quantum-resistant cryptography.

Q

Quantum Advantage: The goal of quantum computers to outperform classical computers.

Quantum Algorithm: A step-by-step procedure, using the principles of quantum mechanics, to solve a problem or perform a task.

Quantum Amplitude Estimation: An algorithm to determine the probability of measuring a specific outcome in a stochastic system.

Quantum Annealing: A quantum algorithm for solving optimization problems by finding low-energy states of a system.

Quantum Circuits: The basic structure for processing quantum information, consisting of qubits and quantum gates.

Quantum Communications: A technology using devices that transmit un-hackable information.

Quantum Computers: Devices that use quantum mechanics for computation.

Quantum Cryptography: An encryption method that uses quantum mechanical properties to secure data.

Quantum Entanglement: A phenomenon where quantum particles become interconnected and the state of one instantly influences the state of another.

Quantum Error Correction: A set of techniques to protect quantum information from errors due to decoherence and other quantum noise.
Quantum Fourier Transform: A linear transformation used in many quantum algorithms to map quantum states and amplitudes.
Quantum Gates: Physical operations performed on qubits to change their states, analogous to logic gates in computer science.
Quantum Inspired: Use of computational methods inspired by the principles of quantum mechanics.
Quantum Interference: A quantum mechanics phenomenon where the probability amplitudes of quantum states overlap and combine.
Quantum Key Distribution: A secure communication method that implements a cryptographic protocol through quantum mechanics to distribute and share keys.
Quantum Mechanics: A physical theory that describes the physical properties of nature at the scale of atoms and subatomic particles.
Quantum Noise: A measured vibration that can lead to inaccuracy.
Quantum Parallelism: The possibility of performing a large number of calculations simultaneously by applying quantum properties.
Quantum Processing Unit: The core component of a quantum computer, and it is the chip where the computation takes place.
Quantum Random Number Generators: Hardware components used for generating secure and unpredictable codes used for cryptography.
Quantum Secure Cryptography: Cryptographic techniques designed to be secure against quantum computing attacks.
Quantum Sensors: Devices that detect physical properties with high sensitivity.
Quantum Superposition: A principle where a quantum system can be in multiple states at the same time.
Quantum Support Vector Machine: Quantum version of machine learning algorithms that maps classical data into a high-dimensional quantum feature space.
Quantum Teleportation: A process of transferring quantum information from one location to another, using entanglement.

Qubit: The basic unit of quantum information, analogous to a bit in classical computing.

R

Reservoir: An oil and gas reservoir is a rock formation in which oil and natural gas have accumulated and are trapped in connected pore spaces of rock.

Rivest-Shamir-Adleman: Public-key cryptosystem used for secure data transmission.

S

Schrödinger Equation: Fundamental equation in quantum mechanics that describes how the quantum state of a physical system changes over time measuring energy levels, named after the physicist Erwin Schrödinger.

Speedup: Process that reduces the number of steps to solve a problem either significantly (exponential) or slightly (polynomial).

Statistical Process Control: The use of statistical techniques to control a process or production method to help improve quality in performance.

Sustainable Development Goals: The 17 world goals part of the 2030 Agenda adopted by all United Nations members in 2015 that highlight the connections between the environmental, social, and economic aspects of sustainable development.

Supply Chain: The network of individuals, organizations, resources, activities, and technology involved in the creation and sale of a product.

T

Technology Adoption Cycle: A sociological model that describes the acceptance of a new product based on the demographic and psychological characteristics.

Thermodynamics: The relationship between heat, work, temperature, and energy.

Threshold Theorem: A framework for quantifying the fault tolerance level to effectively perform quantum computations.

Technology Readiness Level: A scale that assesses the maturity of technologies, ranging from initial concepts to fully operational systems.

Tensor Product: A mathematical operation combining two or more vectors or matrices, used in quantum circuits to combine qubits with quantum operators.

Traveling Salesman Problem: A classic problem in a combinatorial optimization problem that seeks to find the shortest path to visit different cities.

Tunneling: A quantum mechanical phenomenon where a particle passes through a potential barrier despite having less energy than the height of the barrier.

Turbines: Machines for producing continuous power in which a wheel or rotor, fitted with vanes, is made to revolve by a fast-moving flow of different fluids.

U

Unit Commitment Problem: A problem that aims to minimize the total cost of electric power generation in a specific period, balancing and scheduling the generating units.

Unitary Evolution: Refers to the evolution of a time-dependent quantum system that preserves the norm of the state vector, ensuring probabilities sum to one.

Uncertainty Principle: Certain pairs of physical properties, like position and momentum, cannot both be precisely measured at the same time.

V

Vehicle Routing: A problem that seeks to find optimal routes for multiple vehicles visiting a set of locations.

Variational Quantum Algorithms: Hybrid algorithms that use both classical and quantum computing resources to find approximate solutions to problems.

Acronyms

AA	amplitude amplification
AES	advanced encryption standard
AI	artificial intelligence
ATM	automated teller machine
BCG	Boston Consulting Group
CAGR	compound annual growth rate
CCUS	carbon capture, utilization, and storage
CEO	chief executive officer
CISA	Cybersecurity and Infrastructure Security Agency
CPU	central processing unit
CT	computed tomography
ECC	Elliptic Curve Cryptography
ETSI	European Telecommunications Standards Institute
FAA	Federal Aviation Administration
FTQC	fault-tolerant quantum computer
GDP	gross domestic product
GPS	global positioning system
HHL	Harrow–Hassidim–Lloyd
ICT	information and communications technology
IEEE	Institute of Electrical and Electronics Engineers
IMF	International Monetary Fund
IP	internet protocol; intellectual property
IP-VPN	internet protocol-virtual private network
IROPs	irregular operations
ISO	International Organization for Standardization
IT	information technology
ITU	International Telecommunications Union
JQI	Joint Quantum Institute
KEM	key encapsulation mechanism

KYC	know your customer
MIMO	multiple-input, multiple-output
mK	millikelvin
MPLS	multiprotocol label switching
MRI	magnetic resonance imaging
MTBF	mean time between failure
NSA	National Security Agency
NASA	National Aeronautics and Space Administration
NGB	National Guard Bureau
NISQ	noisy intermediate-scale quantum
NIST US	Department of Commerce's National Institute of Standards and Technology
OECD	Organization for Economic Co-operation and Development
OEM	original equipment manufacturer
OPEX	operational expenditures
PET	positron emission tomography
PPPs	plant protection pesticides
PQC	post-quantum cryptography
QAE	quantum amplitude estimation
QAOA	quantum approximate optimization algorithm
QDE	quantum differential equation
QEC	quantum error correction
QED-C	quantum economic development consortium
QFT	quantum Fourier transform
QML	quantum machine learning
QNN	quantum neural network
QPE	quantum phase estimation
QPU	quantum processing unit
QUBO	quadratic unconstrained binary optimization
QRNGs	quantum random number generators
QSVM	quantum support vector machine
PII	personally identifiable information
R&D	research and development
RAN	radio access network
RSA	Rivest-Shamir-Adleman
SDG	sustainable development goals
SDK	software development kit
SD-WAN	software-defined wide area network
SPC	statistical process control

STSM	senior technical staff member
TRL	technology readiness level
TSP	traveling salesman problem
UC3M	Universidad Carlos III de Madrid
UCP	unit commitment problem
VQA	variational quantum algorithms
VQE	variational quantum eigensolver
VR	vehicle routing
UN	United Nations
WEF	World Economic Forum

Introduction

Welcome to *Quantum Computing Strategy: Foundations and Applicability*, a journey into one of the most enigmatic and transformative technologies of our age. In these pages, we will explore the world of quantum computing, a field that challenges our traditional notions of computation and information processing. This book doesn't require advanced mathematical or physics knowledge. It is written for a broad audience, including technology enthusiasts, professionals exploring quantum computing applications in their fields, and policymakers seeking to understand the strategic importance of this emerging technology.

This book covers all necessary technical foundations for understanding what quantum computing is; contextualizes it in the larger quantum technology scheme; explains its key underlying algorithms; defines the problem areas that will benefit most from this technology; and analyzes how to choose an appropriate problem to be solved with this technology, how to approach its development, how to link problem areas with industrial use cases, how to work in a hybrid quantum–classical environment, and how to integrate quantum computing with existing technologies.

The target audience for this book is curious professionals who evaluate technologies and are considering quantum computing, listed below alphabetically:

- Anyone interested in gaining an intuition on how quantum computing works, its use case potential, and an overview of the whole ecosystem.

DOI: 10.1201/9781003302674-1

- Board members who want to understand tech implications and opportunities.
- Business leads who need to understand the benefits of quantum technology.
- DevOps architects who define how to integrate technologies.
- Financial advisors making decisions on high-tech investments.
- Graduate students who want to learn quantum computing.
- Instructors who need to teach this topic to non-STEM students.
- Journalists specialized in technology topics and the latest innovations.
- Product owners who need to determine the applicability of use cases.
- Regulators and policymakers who need to understand the power of quantum technology to assess how to regulate it.
- Software developers who need to obtain a technology and use case applicability overview.

Some of the concepts covered in this book are the peculiar properties of quantum bits (qubits), how they can store multiple states (superposition), how they interact through highly correlated behavior (entanglement), how the algorithms are different from existing ones, how they outperform existing computational systems, and how to categorize the type of problems that they solve including quantum security, the other side of the coin of the technology use, and finally a guide on technology adoption roadmaps.

Although this book defines quantum technologies in general, it is focused on quantum computing and on some of the applicability use cases that industrial corporations are testing. Corporations are thinking about problems that are difficult to solve, cannot be solved quickly, or cannot be properly approached, and this is what motivates them to test quantum computing.

Some examples of quantum computing applicability for business areas covered in this book include the following:

- Aerospace companies are testing air analysis when flying, monitoring operational uncertainties during flight, and material design.

- Agriculture farmers are testing how to improve the quantity and quality of crops and become more sustainable.
- Automotive companies are testing applications in battery design, routing, and vehicle production process.
- Chemical companies are testing catalytic reactions, electrochemical effects, and correlated systems in computational chemistry.
- Energy companies are testing energy resource optimization, reduction of carbon emissions, and management of power generation networks.
- Financial entities are testing applications for risk analysis in portfolio management, product pricing, and fraud detection.
- Government agencies are testing how to enhance country security, city management, and sustainability.
- Healthcare pharmaceutical companies are testing how to improve patient diagnosis and medical treatments.
- Manufacturing companies are testing how to optimize production lines and development of new materials.
- Supply chain and logistics companies are testing how to optimize demand and transportation management.
- Telecom companies are testing how to improve networks from planning to operations to handling multiple devices.

This book provides insights into how businesses, governments, and individuals can prepare for the upcoming quantum era, highlighting the challenges and opportunities that lie ahead. The purpose of this book is twofold.

First, it serves as a primer on the core principles that underpin quantum computing, covered in the first section. Second, this book explores the strategic implications of quantum computing on different industries, covered in the second section.

This book is intended for those who want to understand the technological benefits of quantum computing, the problems it solves, the areas it affects, and its impact on current operations. The goal of this book is

to serve as a "handbook" and as a guide and reference. As such, it can be read sequentially, building knowledge from chapter to chapter, or as a reference guide, where the reader can consult a specific section to understand a problem described for a particular field or industry, the applicability of a use case, how a specific algorithm works, or a detail about the technology stack.

As you turn these pages, I invite you to join me on a journey of discovery, exploring the strange yet fascinating world of quantum computing. It is a journey that will not only expand your understanding but also challenge you to think about the future of technology and its impact on our world.

References

This section provides some resources that can be useful for subject matter experts, researchers, developers, and students, both from the scientific and industry world.

“2020 Airbus Quantum Computing Challenge” Airbus web. 2020 [WEB POST]. https://www.airbus.com/en/innovation/disruptive-concepts/quantum-technologies/2020-airbus-quantum-computing-challenge

“A BBVA and Zapata Computing Study Shows the Potential of Quantum Computing for Derivative Calculations” BBVA Website. 2021 [WEB POST]. https://www.bbva.com/en/innovation/a-bbva-and-zapata-computing-study-shows-the-potential-of-quantum-computing-for-derivative-calculations/

“A Biological Sequence Comparison Algorithm Using Quantum Computers” Nature. 2023 [RESEARCH PAPER]. https://arxiv.org/pdf/2303.13608.pdf

“A Comprehensive Review of Quantum Random Number Generators: Concepts, Classification and the Origin of Randomness” by Vaisakh Mannalath et al. Springer Link. 2023 [RESEARCH PAPER]. https://link.springer.com/article/10.1007/s11128-023-04175-y

“A Conceptual Architecture for a Quantum-HPC Middleware” by IEEE. 2023 [RESEARCH PAPER]. https://arxiv.org/pdf/2308.06608.pdf

“A Fast Quantum Mechanical Algorithm for Database Search” by Lov K. Grover. ACM. 1996 [RESEARCH PAPER]. https://dl.acm.org/doi/pdf/10.1145/237814.237866

“A Fast Quantum Mechanical Algorithm for Database Search” by Lov K. Grover. ACM. 1996 [RESEARCH PAPER]. https://dl.acm.org/doi/pdf/10.1145/237814.237866

“A Fertilizer Revolution is on the Horizon” by Alberta Farmer Express. 2021 [WEB POST]. https://www.albertafarmexpress.ca/news/a-fertilizer-revolution-is-on-the-horizon/

“A Perspective on Protein Structure Prediction using Quantum Computers” by Hakan Doga et al. Inspire High-Energy Physics. 2023 [RESEARCH PAPER]. https://arxiv.org/pdf/2312.00875.pdf

“A Quantum Computing Approach for the Unit Commitment Problem” Springer. 2022 [RESEARCH PAPER]. https://arxiv.org/pdf/2212.06480.pdf

“A Quantum-Inspired Approach to De-Novo Drug Design” Fujitsu. [WHITE PAPER]. https://www.fujitsu.com/fi/imagesgig5/Healthcare-Assets-Whitepaper.pdf

“A Quantum–Quantum Metropolis Algorithm” by Man-Hong Yung and Alán Aspuru-Guzik. PNAS. 2012 [RESEARCH PAPER]. https://www.pnas.org/doi/10.1073/pnas.1111758109

“A Quantum–Quantum Metropolis Algorithm” by Man-Hong Yung and Alán Aspuru-Guzik. PNAS. 2012 [RESEARCH PAPER]. https://www.pnas.org/doi/10.1073/pnas.1111758109

“A Short Introduction to Topological Quantum Computation” SciPostPhys. 2017 [RESEARCH PAPER]. https://scipost.org/SciPostPhys.3.3.021/pdf

“A Step-by-Step HHL Algorithm Walkthrough to Enhance Understanding of Critical Quantum Computing Concepts” by Anika Zaman et al. arXiv. 2023. https://arxiv.org/pdf/2108.09004.pdf

“Accuracy and Resource Estimations for Quantum Chemistry on a Near-term Quantum Computer” by Michael Kühn, Peter Deglmann, and Horst Weiß. ACS. 2018 [RESEARCH PAPER]. https://arxiv.org/pdf/1812.06814.pdf

“Aerospace: Economic Growth” by Aviation Benefits Beyond Borders Website [WEB POST]. https://aviationbenefits.org/economic-growth/

“Aerospace”: https://www.britannica.com/technology/aerospace-industry

“Agriculture and Food” World Bank website. [WEB POST]. https://www.worldbank.org/en/topic/agriculture/overview

“Agriculture”: https://www.nationalgeographic.org/encyclopedia/agriculture/

“Air Force Research Laboratory Partners With QC Ware to Apply Quantum Machine Learning in Identifying Flight Patterns of Unmanned Aircraft” QC Ware. 2021 [WEB POST]. https://www.qcware.com/news/air-force-research-laboratory-partners-with-qc-ware-to-apply-quantum-machine-learning-in-identifying-flight-patterns-of-unmanned-aircraft

“Airbus Gets Aerodynamic With Quantum Computing” The Next Platform. 2019 [WEB POST]. https://www.nextplatform.com/2019/01/24/airbus-gets-aerodynamic-with-quantum-computing/

“An Introduction to Resource Estimation” 2023 [WEB POST]. https://learn.microsoft.com/en-us/azure/quantum/intro-to-resource-estimation

“Analyzing the Economic Impacts of Telecommunications” Utilities One website. 2023 [WEB POST]. Utilities One web site, accessed on September 2023.

“Application Overview of Quantum Computing for Gas Turbine Design and Optimization” by Aurthur Vimalachandran Thomas Jayachandran. IntechOpen Journals. 2022 [RESEARCH ARTICLE]. https://www.intechopen.com/journals/1/articles/106
“Application-Oriented Performance Benchmarks for Quantum Computing” by Thomas Lubinski et al. IEEE. 2023 [RESEARCH PAPER]. https://ieeexplore.ieee.org/document/10061574
“Applying Quantum Algorithms to Satellite Mission Planning Optimization” Tech UK. 2023 [WEB POST]. https://www.techuk.org/resource/applying-quantum-algorithms-to-satellite-mission-planning-optimization-terra-quantum-and-thales-group-unlocked-new-revenue-potential.html
“Are Quantum Computers the Future of Genome Analysis?” Osaka University. 2023 [WEB POST]. https://resou.osaka-u.ac.jp/en/research/2023/20230726_1
“Are Quantum Computers the Future of Genome Analysis?” PhysOrg. 2023 [RESEARCH PAPER]. https://phys.org/news/2023-11-quantum-gene-relationships.html
“Assessing Requirements to Scale to Practical Quantum Advantage” by Michael E. Beverland . arXiv. 2022 [RESEARCH PAPER]. https://arxiv.org/abs/2211.07629.pdf
“Automakers Like BMW Are Becoming Quantum Computing’s Early Adopters” Tech Monitor 30. 2022 [WEB POST]. https://techmonitor.ai/technology/emerging-technology/quantum-computing-automotive-bmw-pasqal
“Automotive Industry Worldwide – Statistics & Facts” Statista. [WEB POST]. https://www.statista.com/topics/1487/automotive-industry/#topicOverview
“Automotive”: https://www.britannica.com/technology/automotive-industry
“Average Profit Margin for Telecommunications Agency: Overview” Investopedia. 2022 [WEB POST]. https://www.investopedia.com/ask/answers/060215/what-average-profit-margin-company-telecommunications-sector.asp
“AWS, Partners Report Successful Quantum Key Distribution Trial in Singapore” by John Russell HPC Wire. 2023 [PRESS RELEASE]. https://www.hpcwire.com/2023/03/06/aws-partners-report-successful-quantum-key-distribution-trial-in-singapore/
“Azure Quantum Elements Aims to Compress 250 Years of Chemistry into the Next 25” Microsoft Website. 2023 [WEB POST]. https://news.microsoft.com/source/features/innovation/azure-quantum-elements-chemistry-materials-science/
“Banco Santander, BBVA and CaixaBank Team Up to Tackle Fraud with FrauDfense” Fintech Futures. 2023 [WEB POST]. https://www.fintechfutures.com/2023/07/banco-santander-bbva-and-caixabank-unify-fraud-defences-with-fraudfense/
“BASF Collaborates with PASQAL to Predict Weather Patterns” Pasqal. 2022 [PRESS RELEASE]. https://www.pasqal.com/articles/basf-collaborates-with-pasqal-to-predict-weather-patterns
“Bayer Quantum-inspired Computing’s Potential to Raise Yields” Fujitsu. 2022 [CASE STUDY].
“BBVA and Multiverse Showcase How Quantum Computing Could Help Optimize Investment Portfolio Management” BBVA. 2022 [WEB POST]. https://www.bbva.com/en/bbva-and-multiverse-showcase-how-quantum-computing-could-help-optimize-investment-portfolio-management/
“Beyond the Hype: A Critical Look at Quantum Computing’s Potential for Business and Society in Asia-Pacific” EY Website [WHITE PAPER]. https://assets.ey.com/content/dam/ey-sites/ey-com/en_gl/topics/financial-services-asia-pacific/beyond-the-hype-a-critical-look-at-quantum-computings-potential-for-business-and-society-in-asia-pacific.pdf; https://www.eneos-innovation.co.jp/english/newsroom/20200421
“Blueprint for a Scalable Photonic Fault-Tolerant Quantum Computer” by J. Eli Bourassa, et al. Quantum Open Journal. 2021 [RESEARCH PAPER]. https://quantum-journal.org/papers/q-2021-02-04-392/
“BMO Financial Group and Scotiabank Partner with Xanadu on Quantum Computing Speedups for Trading Products” Xanadu. 2019 [PRESS RELEASE]. https://www.prnewswire.com/news-releases/bmo-financial-group-and-scotiabank-partner-with-xanadu-on-quantum-computing-speedups-for-trading-products-300904106.html
“BMW Developing More Durable, Safer and Less Expensive Next Generation Batteries” [CASE STUDY]. https://www.pasqal.com/industry/mobility/battery-modelling
“Bosch to Build Quantum Digital Twin of Madrid Factory” Yole Group. 2022 [WEB POST]. https://www.yolegroup.com/industry-news/bosch-to-build-quantum-digital-twin-of-madrid-factory/

“BP Joins the IBM Quantum Network to Advance Use of Quantum Computing in Energy” Green Car Congress. 2021 [WEB POST]. https://www.greencarcongress.com/2021/02/20210222-bp.html
“Brief History of Quantum Cryptography: A Personal Perspective” by Gilles Brassard. arXiv. 2006 [RESEARCH PAPER]. https://arxiv.org/abs/quant-ph/0604072
“Can Quantum Computers Bring an End to Corrosion?” IBM Research Website. 2023 [WEB POST]. https://research.ibm.com/blog/boeing-quantum-corrosion
“Carbon-capture Technology Could Benefit from Quantum Computing” Physics World. 2023 [WEB POST]. https://physicsworld.com/a/carbon-capture-technology-could-benefit-from-quantum-computing/
“Chemical Industry Contributes $5.7 Trillion to Global GDP and Supports 120 Million Jobs” International Council of Chemical Associations (ICCA) Website. 2019 [REPORT]. https://icca-chem.org/news/chemical-industry-contributes-5-7-trillion-to-global-gdp-and-supports-120-million-jobs-new-report-shows/
“Chemicals”: https://www.britannica.com/technology/chemical-industry
“Citi and Classiq Advance Quantum Solutions for Portfolio Optimization Using Amazon Braket” by Yoram Avidan et al. AWS Website. 2024 [BLOG POST]. https://aws.amazon.com/blogs/quantum-computing/citi-and-classiq-advance-quantum-solutions-for-portfolio-optimization/
“Cold Atom Qubits” by Dmitry Solenov and Dmitry Mozyrsky. American Scientific Publishers. 2010 [RESEARCH PAPER]. https://arxiv.org/abs/1005.5487
“Combining the QAOA and HHL Algorithm to Achieve a Substantial Quantum Speedup for the Unit Commitment Problem” by Jonas Stein et al. 2023 [RESEARCH PAPER]. https://arxiv.org/pdf/2305.08482.pdf
“Communication Theory of Secrecy Systems” by Claude E. Shannon. Bell System Technical Journal, 1949 [RESEARCH PAPER]. https://ieeexplore.ieee.org/document/6769090
“Conjugate Coding” by Stephen Weisner . ACM SIGACT, 1983 [RESEARCH PAPER]. https://dl.acm.org/doi/10.1145/1008908.1008920
“Cooling Quantum Computers: Keeping Your Qubits Stable Requires Some of the Most Extreme Cooling Equipment Around” by Sebastian Moss . DCD. 2021 [WEB POST]. https://www.datacenterdynamics.com/en/analysis/cooling-quantum-computers/
“Could Quantum Computing Clean Up the Ongoing Air Travel Mess?” Inside Quantum Technology. 2022 [WEB ARTICLE]. https://www.insidequantumtechnology.com/news-archive/could-quantum-computing-clean-up-the-ongoing-air-travel-mess/;
“Could Quantum Computing Make Our Energy Grid Sustainable?” Tech Monitor. 2023 [WEB POST]. https://techmonitor.ai/hardware/quantum/can-quantum-computing-make-the-energy-grid-sustainable
“Crossing the Chasm – Geoffrey Moore” Strategies for Influence. [WEB POST]. https://strategiesforinfluence.com/crossing-the-chasm-geoffrey-moore/
“Delta Partners with IBM to Explore Quantum Computing – An Airline Industry First” IBM. 2020 [WEB POST]. https://newsroom.ibm.com/2020-01-08-Delta-Partners-with-IBM-to-Explore-Quantum-Computing-an-Airline-Industry-First
“Demystifying Cybersecurity: Post-Quantum Security” ISEC7. 2023 [WEB POST]. https://blog.isec7.com/en/demystifying-cybersecurity-post-quantum-security-part-1
https://blog.isec7.com/en/demystifying-cybersecurity-post-quantum-security-part-2
“Digital Technologies to Achieve the UN SDGs” by ITU. 2021 [WEB POST]. https://www.itu.int/en/mediacentre/backgrounders/Pages/icts-to-achieve-the-united-nations-sustainable-development-goals.aspx
“Digitized Counterdiabatic Quantum Algorithm for Protein Folding” by APS. 2023 [RESEARCH PAPER]. https://journals.aps.org/prapplied/abstract/10.1103/PhysRevApplied.20.014024
“DLR and NASA Are Jointly Developing a Software Package for Quantum Computers” DLR. 2022 [WEB POST]. https://www.dlr.de/en/latest/news/2022/01/20220303_dlr-and-nasa-developing-software-package-for-quantum-computers
“E.ON to Manage Decentralised Energy System Using Quantum Computing” Smart Energy International. 2021 [WEB POST]. https://www.smart-energy.com/industry-sectors/digitalisation/e-on-to-manage-decentralised-energy-system-using-quantum-computing/
“Efficient Quantum Circuits for Schur and Clebsch-Gordan Transforms,” by D. Bacon, I. L. Chuang, and A. W. Harrow, Phys. Rev. Lett. 97, 170502 (2006); “Efficient Compression of

Quantum Information," by M. Plesch and V. Buzek , Phys. Rev. A 81, 032317 (2010).
"Electronics": https://www.investopedia.com/ask/answers/042915/what-electronics-sector.asp
"Elliptic Curve Cryptography: What Is It? How Does It Work? Keyfactor [WEB POST]. https://www.keyfactor.com/blog/elliptic-curve-cryptography-what-is-it-how-does-it-work/
"Elucidating Reaction Mechanisms on Quantum Computers" by Markus Reiher et al. PNAS. 2017 [RESEARCH PAPER]. https://www.pnas.org/doi/10.1073/pnas.1619152114
"ENEOS Leads the Way to Sustainable Hydrogen Fuel by Validating Computing Algorithms Using Azure Quantum" Microsoft. 2022 [CASE STUDY]. https://customers.microsoft.com/en-us/story/1508526643598641548-eneos-energy-azure-quantum
"Energy": https://www.investopedia.com/terms/e/energy_sector.asp
"Eni and PASQAL Together to Develop Quantum Solutions for the Energy Sector" Eni. 2022 [USE CASE]. https://www.eni.com/en-IT/media/press-release/2022/11/eni-pasqal-together-develop-quantum-solutions-energy-sector.html
"Entanglement and Quantum Information", by P.G. Kwiat , D.F.V. James. Science Direct. 2023 [WEB POST]. https://www.sciencedirect.com/topics/engineering/quantum-teleportation
"Error Mitigation Extends the Computational Reach of a Noisy Quantum Processor" by Abhinav Kandala et al. Nature. 2019. https://www.nature.com/articles/s41586-019-1040-7
"Explainer: What Is Quantum Communication?" by Martin Giles . MIT Technology Review. 2019 [WEB POST]. https://www.technologyreview.com/2019/02/14/103409/what-is-quantum-communications/
"Exploiting Fermion Number in Factorized Decompositions of the Electronic Structure Hamiltonian" by Sam McArdle. 2022 [RESEARCH PAPER]. https://arxiv.org/pdf/2107.07238.pdf
"Explore Quantum. Quantum Error Correction" Quantum Microsoft Web. [WEB POST]. https://quantum.microsoft.com/en-us/explore/concepts/quantum-error-correction#:~:text=Current%20state%2Dof%2Dthe%2D,a%201%20or%20vice%20versa
"Exponential Algorithmic Speedup by Quantum Walk" by Andrew M. Childs et al. arXiv. 2002 [RESEARCH PAPER]. https://arxiv.org/pdf/quant-ph/0209131.pdf
"ExxonMobil & IBM Explore Quantum Algorithms to Solve Routing Formulations" IBM Medium. 2021 [WEB POST]. https://ibm-research.medium.com/exxonmobil-ibm-scientists-explore-state-of-art-quantum-algorithms-to-solve-routing-formulations-e7ce39f8741c
"ExxonMobil, NCSA, Cray Scale Reservoir Simulation to 700,000+ Processors" HPCwire. 2017 [PRESS RELEASE]. https://www.hpcwire.com/2017/02/17/exxonmobil-ncsa-cray-scale-reservoir-simulation-700000-processors/
"Fault-tolerant Resource Estimate for Quantum Chemical Simulations: Case study on Li-ion Battery Electrolyte Molecules" by Isaac H. Kim. APS. 2022 [RESEARCH PAPER]. https://journals.aps.org/prresearch/abstract/10.1103/PhysRevResearch.4.023019
"FedEx's Investments in Quantum Computing: What's on the Horizon" Techbuillon. 2023 [WEB ARTICLE]. https://techbullion.com/fedexs-investments-in-quantum-computing-whats-on-the-horizon/
"Fertilizer and Other Quantum Computer Chemistry" Quantum Flagship. [WEB POST]. https://qt.eu/applications/fertilizer-and-other-quantum-computer-chemistry
"Finance": https://www.investopedia.com/terms/f/financial_sector.asp
"Financial Services: Sizing the Sector in the Global Economy" Investopedia. 2021 [WEB POST]. https://www.investopedia.com/ask/answers/030515/what-percentage-global-economy-comprised-financial-services-sector.asp
"Finding flows of a Navier–Stokes Fluid through Quantum Computing" by Frank Gaitan. Nature. 2020 [RESEARCH PAPER]. https://www.nature.com/articles/s41534-020-00291-0
"Ford Enlists Quantum Computing in EV Battery Materials Hunt" EE Times Europe. 2023 [WEB POST]. https://www.eetimes.eu/ford-enlists-quantum-computing-in-ev-battery-materials-hunt/
"Ford, Microsoft Partner to Reduce Congestion with 'Quantum-inspired' Tech" Ford Media. 2019 [PRESS RELEASE]. https://media.ford.com/content/fordmedia/fna/us/en/news/2019/12/10/ford-exploring-quantum-world-with-microsoft.html
"Formulating and Solving Routing Problems on Quantum Computers" by Stuart Harwood et al. IEEE. 2021 [RESEARCH PAPER]. https://ieeexplore.ieee.org/document/9314905
"Fuel Consumption Minimization of Transport Aircraft Using Real-Coded Genetic Algorithm" Sage Journals. 2017 [RESEARCH PAPER]. https://journals.sagepub.com/doi/full/10.1177/0954410017705899

“Global Pharmaceutical Industry – Statistics & Facts” Statista Key Insights [WEB POST]. https://www.statista.com/topics/1764/global-pharmaceutical-industry/#topicOverview
“Global Supply Chain Management (SCM) Market to Grow from $28.9 Billion to $45.2 Billion by 2027 at a CAGR of 9.4%” Research and Markets. 2022 [PRES RELEASE]. https://www.globenewswire.com/en/news-release/2022/06/29/2471248/28124/en/Global-Supply-Chain-Management-SCM-Market-to-Grow-from-28-9-Billion-to-45-2-Billion-by-2027-at-a-CAGR-of-9-4.html
“Government”: https://www.britannica.com/summary/government
“GSK on Quantum Use Cases in Pharmaceuticals” IoT World Today. 2022 [VIDEO]. https://www.quantumbusinessnews.com/applications/video-gsk-on-quantum-computing-use-cases-for-the-pharmaceutical-industry
“How BMW Can Maximize Its Supply Chain Efficiency with Quantum” Honeywell. [News Post]. https://www.honeywell.com/us/en/news/2021/01/exploring-supply-chain-solutions-with-quantum-computing
“How Many People Work for the Federal Government?” USA Facts. 2023 [WEB POST]. https://usafacts.org/articles/how-many-people-work-for-the-federal-government/#:~:text=As%20of%20September%202023%2C%20there,increasing%20again%20in%20the%201950s.
“How Much Revenue Has the U.S. Government Collected This Year?” FiscalData. [WEB POST]. https://fiscaldata.treasury.gov/americas-finance-guide/government-revenue/
“How NASA Utilizes Quantum Computing for Weather Forecasting” Tech Buillon. 2023 [WEB POST]. https://techbullion.com/how-nasa-utilizes-quantum-computing-for-weather-forecasting/
“How Quantum Computers Could Help Design Airplanes” IBM. 2023 [CASE STUDY]. https://research.ibm.com/blog/boeing-case-study
“How to Factor 2048 Bit RSA Integers in 8 Hours Using 20 Million Noisy Qubits” by Craig Gidney and Martin Ekerå. arXiv. 2019 [RESEARCH PAPER]. https://arxiv.org/abs/1905.09749
“How VW, Bosch, Ford, Daimler Aim to Gain from Quantum Computing” Europe Autonews. 2019 [WEB POST]. https://europe.autonews.com/automakers/how-vw-bosch-ford-daimler-aim-gain-quantum-computing
“Hybrid Quantum-classical Algorithms in the Noisy Intermediate-scale Quantum Era and beyond” by Adam Callison and Nicholas Chancellor. Phys. Rev. 2022 [RESEARCH PAPER]. https://journals.aps.org/pra/abstract/10.1103/PhysRevA.106.010101
“Industry vs. Sector: What’s the Difference?”: https://www.investopedia.com/ask/answers/05/industrysector.asp
“Inside PayPal’s Partnership with IBM to Use Quantum Computing to Improve How It Detects Fraud and Underwrites” Business Insider. 2022 [WEB POST]. https://www.businessinsider.com/paypal-quantum-computing-fraud-prevention-lending-machine-learning-payments-tech-2022-1
“IonQ and Hyundai Motor Partner to Use Quantum Computing to Advance Effectiveness of Next-Gen Batteries” Hyundai. 2022 [PRESS RELEASE]. https://www.hyundai.news/eu/articles/press-releases/ionq-and-hyundai-partner-to-use-quantum-computing-to-advance-effectiveness-of-next-gen-batteries.html
“IonQ, Airbus Sign Agreement to Collaborate on Aircraft Loading Project Using Quantum Computing” IonQ Web. 2022 [WEB POST]. https://ionq.com/news/august-18-2022-ionq-2022-airbus
“Jülich Quantum Computer Solves Protein Puzzle” Julich. 2023 [WEB POST]. https://www.fz-juelich.de/en/news/archive/press-release/2023/julich-quantum-computer-solves-protein-puzzle
“Linear Algebra for Quantum Computing” Microsoft. 2023 [WEB POST]. https://learn.microsoft.com/en-us/azure/quantum/overview-algebra-for-quantum-computing#vectors-and-matrices-in-quantum-computing
“Low Depth Amplitude Estimation on a Trapped Ion Quantum Computer” by Tudor Giurgica-Tiron. arXiv. 2021 [RESEARCH PAPER]. https://arxiv.org/abs/2109.09685
“Medical Image Classification via Quantum Neural Networks” by Natansh Mathur. arXiv, 2021 [RESEARCH PAPER]. https://arxiv.org/abs/2109.01831
“Mercedes-Benz Bets on Quantum to Craft the Future of Electric Vehicles” IBM. [CASE STUDY]. https://www.ibm.com/case-studies/daimler
“Middleware for Quantum: An Orchestration of Hybrid Quantum-Classical Systems” IEEE. 2023 [CONFERENCE PAPER]. https://ieeexplore.ieee.org/document/10234290

“Migration to Post-Quantum Cryptography” NIST. 2023 [PRESENTATION]. https://www.nccoe.nist.gov/crypto-agility-considerations-migrating-post-quantum-cryptographic-algorithms
“Modelling Carbon Capture on Metal-Organic Frameworks with Quantum Computing” Springer. 2022 [RESEARCH PAPER]. https://epjquantumtechnology.springeropen.com/articles/10.1140/epjqt/s40507-022-00155-w
“Moving the World’s Energy Products across the Globe Is a Complex Puzzle That Could Benefit from a Quantum Solution” IBM website. 2020 [CASE STUDY]. https://www.ibm.com/case-studies/exxonmobil
“NIST Researchers Help Design a Prototype Quantum Computer” by NIST. 2023 [WEB POST]. https://www.nist.gov/news-events/news/2023/12/nist-researchers-help-design-prototype-quantum-computer
“NIST to Standardize Encryption Algorithms That Can Resist Attack by Quantum Computers” NIST. 2023 [WEB POST]. https://www.nist.gov/news-events/news/2023/08/nist-standardize-encryption-algorithms-can-resist-attack-quantum-computers
“Nomura Launches Joint Research on Quantum Computing” Nomura. 2018 [PRESS RELEASE]. https://www.nomuraholdings.com/news/nr/holdings/20180227/20180227.pdf
“Nvidia Spools Up Quantum Jet-Engine Simulations” by Edd Gent. IEEE. 2023 [RESEARCH PAPER]. https://spectrum.ieee.org/computational-fluid-dynamics-quantum-computer
“NVIDIA, Rolls-Royce and Classiq Announce Quantum Computing Breakthrough for Computational Fluid Dynamics in Jet Engines” Nvidia. 2023 [PRESS RELEASE]. https://nvidianews.nvidia.com/news/nvidia-rolls-royce-and-classiq-announce-quantum-computing-breakthrough-for-computational-fluid-dynamics-in-jet-engines
“Oil and Gas Industry Faces Moment of Truth – and Opportunity to Adapt – as Clean Energy Transitions Advance” International Energy Agency: IEA. 2023 [WEB POST]. https://www.iea.org/news/oil-and-gas-industry-faces-moment-of-truth-and-opportunity-to-adapt-as-clean-energy-transitions-advance
“OKI Uses D-Wave Quantum Computer to Increase Manufacturing Productivity” LaserFocus. 2019 [WEB POST]. https://www.laserfocusworld.com/test-measurement/article/14067894/oki-uses-dwave-quantum-computer-to-increase-manufacturing-productivity
“Optimization of Manufacturing Equipment Layouts in Factories Employing Quantum Computer” Oki. 2020 [WHITE PAPER]. https://www.oki.com/en/otr/2020/n235/pdf/otr-235-R13.pdf
“Option Pricing Using Quantum Computers” by Nikitas Stamatopoulos . Quantum. 2020 [RESEARCH PAPER]. https://arxiv.org/pdf/1905.02666.pdf
“Oxford Joins Consortium to Advance Quantum Drug Discovery with £6.8M Grant from Innovate UK” Oxford University. 2021 [WEB POST]. https://www.ox.ac.uk/news/2021-11-05-oxford-joins-consortium-advance-quantum-drug-discovery-68m-grant-innovate-uk
“Pharma Giant Roche Partners for Quantum Drug Discovery” The Next Platform. 2021 [PRESS RELEASE]. https://www.nextplatform.com/2021/01/28/pharma-giant-roche-partners-for-quantum-drug-discovery/
“Pharma”: https://www.investopedia.com/articles/markets/051316/industry-handbook-pharma-industry.asp
“Picturing the Future with Quantum-Enabled Road Sign Recognition” IonQ. 2023 [CASE STUDY]. https://ionq.com/resources/picturing-the-future-quantum-enabled-road-sign-recognition
“Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer” ACM. 1997 Peter Shor [RESEARCH PAPER]. https://dl.acm.org/doi/10.1137/S0097539795293172
“Practical Quantum Security for Grid Automation: Minimize Cost and Increase Security by Enabling a Grid-compatible, Long-term QKD Solution for Guaranteed Secure Communications” by US Department of Energy. 2013 [RESEARCH PROJECT]. https://www.energy.gov/sites/prod/files/2017/04/f34/ORNL_Practical_Quantum_Security_FactSheet_0.pdf
“Proof of Concept Showed Ability to Detect and Defend against Potential Threats and Eavesdroppers” by Toshiba. 2022 [PRESS RELEASE]. https://news.toshiba.com/press-releases/press-release-details/2022/JPMorgan-Chase-Toshiba-and-Ciena-Build-the-First-Quantum-Key-Distribution-Network-Used-to-Secure-Mission-Critical-Blockchain-Application/default.aspx
“Protein Folding Takes a Step Forward With Quantum Computing” QuantumZeitgeist. 2022 [WEB POST]. https://quantumzeitgeist.com/protein-folding-takes-a-step-forward-with-quantum-

computing/
“Protocol to Identify a Topological Superconducting Phase in a Three-terminal Device” by Dmitry I. Pikulin et al. arXiv. 2021 [RESEARCH PAPER].
“Quandela and EDF Work Together to Use Photonic Quantum Computing to Simulate Hydroelectric Dam Structures” Quandela. 2022 [PRESS RELEASE]. https://www.quandela.com/wp-content/uploads/2022/11/Quandela-and-EDF-collaborate-Press-Release.pdf
“Quantum Algorithm for Linear Systems of Equations” by Aram W. Harrow, Avinatan Hassidim, and Seth Lloyd. APS. 2008 [RESEARCH PAPER]. https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.103.150502
“Quantum Algorithms Applied to Satellite Mission Planning for Earth Observation” by Sam McArdle. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing. 2023 [RESEARCH PAPER]. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10155128
“Quantum Algorithms: A Survey of Applications and End-to-end Complexities” by Alexander M. Dalzell et al. arXiv. 2023 [RESEARCH PAPER]. https://arxiv.org/pdf/2310.03011.pdf
“Quantum Amplitude Amplification and Estimation” by Gilles Brassard et al. arXiv. 2000 [RESEARCH PAPER]. https://arxiv.org/pdf/quant-ph/0005055.pdf
“Quantum and Classical Machine Learning for the Classification of Non-small-cell Lung Cancer Patients” by Siddhant Jain et al. 2020 [RESEARCH PAPER]. https://link.springer.com/article/10.1007/s42452-020-2847-4
“Quantum Chemistry in the Age of Quantum Computing” by Yudong Cao et al. ACS, 2019 [RESEARCH PAPER]. https://pubs.acs.org/doi/10.1021/acs.chemrev.8b00803
“Quantum Communications to Move towards an Ultra-secure Smart Grid” by TECNALIA. 2023 [PRESS RELEASE]. https://www.tecnalia.com/en/press-room/quantum-communications-to-move-towards-an-ultra-secure-smart-grid
“Quantum Complexity Theory” by Ethan Bernstein and Umesh Vazirani. ACM, 1993 [RESEARCH PAPER]. https://dl.acm.org/doi/10.1145/167088.167097
“Quantum Computations with Cold Trapped Ions” by J. I. Cirac and P. Zoller . Phys. Rev., 1995 [RESEARCH PAPER]. https://iontrap.umd.edu/wp-content/uploads/2013/10/Quantum-computations-with-cold-trapped-ions.pdf
“Quantum Computers for Weather and Climate Prediction: The Good, the Bad and the Noisy” American Meteorological Society. 2023 [RESEARCH PAPER]. https://arxiv.org/abs/2210.17460
“Quantum Computers Take Us to New Dimensions” Infineon. [WEB POST]. https://www.infineon.com/cms/en/product/promopages/quantumcomputing/
“Quantum Computing and Genome Sequencing Unlocking Genetic Mysteries” UtiliesOne. 2023 [WEB POST].
“Quantum Computing and Materials Science: A Practical Guide to Applying Quantum Annealing to the Configurational Analysis of Materials” by B. Camino. Appl. Phys., 2023 [RESEARCH ARTICLE]. https://pubs.aip.org/aip/jap/article/133/22/221102/2896017/Quantum-computing-and-materials-science-A
“Quantum Computing and Simulations for Energy Applications” by Hari Paudel. IEEE International Conference on Quantum Computing and Engineering QCE22. 2022 [PRESENTATION]. https://www.osti.gov/servlets/purl/1971602
“Quantum Computing Breakthrough Heralds a New Era of Jet Engine Design” Financial Express. 2023 [WEB POST]. https://www.financialexpress.com/business/defence-a-quantum-computing-breakthrough-heralds-a-new-era-of-jet-engines-3119729/
“Quantum Computing Breakthrough in Simulating Chemical Molecules” 2023 [WEB POST]. https://www.eetimes.eu/quantum-computing-breakthrough-in-simulating-chemical-molecules/
“Quantum Computing Enables Unprecedented Materials Science Simulations” Office of Science. 2021 [WEB POST]. https://www.energy.gov/science/ascr/articles/quantum-computing-enables-unprecedented-materials-science-simulations
“Quantum Computing for Chemical and Biomolecular Product Design” by Martin P. Andersson et al. Science Direct. 2022 [RESEARCH PAPER]. https://www.sciencedirect.com/science/article/pii/S2211339821000861
“Quantum Computing for Chemistry” Kassal Group. [WEB POST]. https://www.kassal.group/research/quantum/
“Quantum Computing for Energy Systems Optimization: Challenges and Opportunities” by Akshay Ajagekar, Fengqi You. arXiv. 2020 [RESEARCH PAPER].

https://arxiv.org/pdf/2003.00254
“Quantum Computing Governance Principles” by WEF. 2022 [INSIGHT REPORT]. https://www3.weforum.org/docs/WEF_Quantum_Computing_2022.pdf
“Quantum Computing in Material Discovery” Infosys. [WEB POST]. https://blogs.infosys.com/emerging-technology-solutions/quantum-computing/quantum-computing-in-material-discovery.html
“Quantum Computing is Becoming Business Ready” BCG. 2023 [REPORT]. https://www.bcg.com/publications/2018/next-decade-quantum-computing-how-play
“Quantum Computing May Create Ethical Risks for Businesses. It’s Time to Prepare” by a Scott Buchholz and Beena Ammanath. Deloitte Insights. 2022 [WEB POST]. https://www2.deloitte.com/us/en/insights/topics/cyber-risk/quantum-computing-ethics-risks.html
“Quantum Computing on Azure | How It Works, What’s Coming, & What You Can Try Today” Microsoft Blog. 2023 [WEB POST]. https://techcommunity.microsoft.com/t5/microsoft-mechanics-blog/quantum-computing-on-azure-how-it-works-what-s-coming-amp-what/ba-p/3762127
“Quantum Computing: A New Threat to Cybersecurity” by Michele Mosca. Global Risk Institute. for Quantum Computing [REPORT]
“Quantum Computing: Boehringer Ingelheim and Google Partner for Pharma R&D” Boehringer-Ingelheim website. 2021 [News Post]. https://www.boehringer-ingelheim.com/press-release/partnering-google-quantum-computing
“Quantum Computing” Rolls Royce. 2023 [WEB POST]. https://www.rolls-royce.com/media/our-stories/discover/2023/trent-xwb-quantum-computing.aspx
“Quantum Data Encoding: A Comparative Analysis of Classical-to-Quantum Mapping Techniques and Their Impact on Machine Learning Accuracy” by Minati Rath and Hema Date. 2023 [RESEARCH PAPER]. https://arxiv.org/pdf/2311.10375.pdf
“Quantum Dots: Everything You Need to Know” EDN. 2019 [WEB POST]. https://www.edn.com/quantum-dots-explained/#google_vignette
“Quantum Encoding: An Overview” Quantum Zeitgeist. 2021 [WEB POST]. https://quantumzeitgeist.com/quantum-encoding-an-overview/
“Quantum Encryption vs. Post-Quantum Cryptography” QuantumXC. 22 March 2022. https://quantumxc.com/blog/quantum-encryption-vs-post-quantum-cryptography-infographic/
“Quantum Machine Learning Beyond Kernel Methods” by Sofiene Jerbi et al. Nature. 2023 [RESEARCH PAPER]. https://www.nature.com/articles/s41467-023-36159-y
“Quantum Machine Learning for Better Image Recognition” QM Ware. 2023 [WEB POST]. https://www.qm-ware.com/use-cases/quantum-machine-learning-automotive/
“Quantum Machine Learning: Benefits and Practical Examples” by Frank Phillipson. CEUR Workshop. 2020 [RESEARCH PAPER]. https://ceur-ws.org/Vol-2561/paper5.pdf
“Quantum Machine Learning” by Jacob Biamonte . Nature. 2017 [RESEARCH PAPER]. https://www.nature.com/articles/nature23474
“Quantum Neuromorphic Computing” by Danijela Marković and Julie Grollier. arXiv. 2020 [RESEARCH PAPER]. https://arxiv.org/abs/2006.15111
“Quantum Pricing with a Smile: Implementation of Local Volatility Model on Quantum Computer” by Kazuya Kaneko. Springer. 2022 [RESEARCH PAPER]. https://epjquantumtechnology.springeropen.com/articles/10.1140/epjqt/s40507-022-00125-2
“Quantum Random Number Generation” by Xiongfeng Ma et al. Nature. 2016 [RESEARCH PAPER]. https://www.nature.com/articles/npjqi201621
“Quantum Random Number Generators” by Miguel Herrero-Collantes and Juan Carlos Garcia-Escartin. arXiv. 2016 [RESEARCH PAPER]. https://arxiv.org/pdf/1604.03304.pdf
“Quantum Speedup of Monte Carlo Methods” by Ashley Montanaro. arXiv. 2017 [RESEARCH PAPER]. https://arxiv.org/pdf/1504.06987.pdf
“Quantum Speedup of the Travelling Salesman Problem for Bounded-degree Graphs” by Alexandra E. Moylett et al. arXiv. 2016 [RESEARCH PAPER]. https://arxiv.org/pdf/1612.06203.pdf
“Quantum Support Vector Machines for Aerodynamic Classification” by Xi-Jun Yuan et al. Intelligent Computing. 2023 [RESEARCH PAPER]. https://spj.science.org/doi/pdf/10.34133/icomputing.0057
“Quantum Technologies: A Potential Game-changer in Aerospace” Airbus. [WEB POST]. https://www.airbus.com/en/innovation/disruptive-concepts/quantum-technologies

“Quantum Technology Monitor” by McKinsey. 2023 [REPORT]. https://www.mckinsey.com/~/media/mckinsey/business%20functions/mckinsey%20digital/our%20insights/quantum%20technology%20sees%20record%20investments%20progress%20on%20talent%20gap/quantum-technology-monitor-april-2023.pdf

“Quantum Technology Strengthens Medical Imaging Methods through Quanvolutional Neural Network” Quantum Zeitgeist. 2022 [WEB POST]. https://quantumzeitgeist.com/quantum-technology-strengthens-medical-imaging-methods-through-quanvolutional-neural-network/

“Quantum Technology: The Second Quantum Revolution” by Jonathan P. Dowling and Gerard J. Milburn. The Royal Society Publishing. 2003 [RESEARCH PAPER]. https://royalsocietypublishing.org/doi/epdf/10.1098/rsta.2003.1227

“Quantum-readiness: Migration to Post-quantum Cryptography” by NIST. 2023 [FACT SHEET]. https://www.nccoe.nist.gov/sites/default/files/2023-08/quantum-readiness-fact-sheet.pdf

“Quantum-Safe Cryptography (QSC)” ETSI. [WEB POST]. https://www.etsi.org/technologies/quantum-safe-cryptography#:~:text=The%20ETSI%20Cyber%20Quantum%20Safe,algorithm%20research%2C%20as%20well%20as

“Qubit Engineering Optimizes Wind Farm Energy Production with Azure Quantum” Microsoft. 2022 [WEB POST]. https://cloudblogs.microsoft.com/quantum/2022/05/17/qubit-engineering-optimizes-wind-farm-energy-production-with-azure-quantum/

“Qubit Engineering Optimizes Wind Farm Energy Production with Azure Quantum” Microsoft. 2022 [WEB POST]. https://cloudblogs.microsoft.com/quantum/2022/05/17/qubit-engineering-optimizes-wind-farm-energy-production-with-azure-quantum/

“Rapid Solutions of Problems by Quantum Computation” by David Deutsch and Richard Jozsa. Proceedings of the Royal Society of London. 1992 [RESEARCH PAPER]. https://royalsocietypublishing.org/doi/10.1098/rspa.1992.0167

“Reducing the Number of Qubits from 2 to log2 () to Solve the Traveling Salesman Problem with Quantum Computers: A Proposal for Demonstrating Quantum Supremacy in the NISQ Era” by Mehdi Ramezani et al. arXiv. 2024 [RESEARCH PAPER]. https://arxiv.org/pdf/2402.18530.pdf

“Resource-Efficient Quantum Algorithm for Protein Folding” by Anton Robert et al. Nature. 2021 [RESEARCH PAPER]. https://www.nature.com/articles/s41534-021-00368-4

“Rigetti Enhances Predictive Weather Modeling with Quantum Machine Learning” Rigetti Computing. 2021 [PRESS RELEASE]. https://www.globenewswire.com/news-release/2021/12/01/2344216/0/en/Rigetti-Enhances-Predictive-Weather-Modeling-with-Quantum-Machine-Learning.html

“Robotics and Computer-Integrated Manufacturing” by Marcelo Luis Ruiz Rodríguez. ScienceDirect. 2022 [RESEARCH PAPER]. https://www.sciencedirect.com/science/article/pii/S0736584522000928

“Roche Quantum Neural Network” QCware. 2021 [CASE STUDY]. https://www.qcware.com/customers/roche

“Rolls-Royce, Riverlane Partner on Quantum Materials Simulation” IoT World Today. 2023 [WEB POST]. https://www.iotworldtoday.com/quantum/rolls-royce-riverlane-partner-on-quantum-materials-simulation

“Secure Quantum Key Distribution with Realistic Devices” by Feihu Xu et al. APS. 2020 [RESEARCH PAPER]. https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.92.025002

“Simulating a Discrete Time Crystal” Medium Qiskit. 2021 [WEB POST]. https://medium.com/qiskit/simulating-a-discrete-time-crystal-using-qiskit-531bda6663b7

“Smart Agriculture Decision Making Scheme Using Quantum Annealing” IEEE International Conference on Quantum Computing and Engineering, 2022 [RESEARCH PAPER]. https://ieeexplore.ieee.org/document/9951228

“Steady Progress in Approaching the Quantum Advantage.” 2024 [REPORT]. https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/steady-progress-in-approaching-the-quantum-advantage

“Superconducting Quantum Bits” by John Clarke and Frank K. Wilhelm. Nature. 2008 [RESEARCH PAPER]. https://www.nature.com/articles/nature07128

“Supply Chain”: https://www.investopedia.com/terms/l/logistics.asp

“Sustainable Development. SDG 17 goals” by UN. [WEB POST]. https://sdgs.un.org/goals

“Technology Quarterly. Here, There and Everywhere” The Economist. [NEWS POST]. https://www.economist.com/news/essays/21717782-quantum-technology-beginning-come-its-own
“Technology Readiness Levels” NASA. 2023 [WEB POST]. https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/
“Telecommunications”: https://www.investopedia.com/ask/answers/070815/what-telecommunications-sector.asp
“Teleporting an Unknown Quantum State via Dual Classical and Einstein-Podolsky-Rosen Channels” by Charles Benett et al. Phys. Rev. Lett. 1993 [RESEARCH PAPER].
“The Case for Quantum and Quantum-Inspired Computing in Financial Services” Fujistu. [WHITE PAPER]. https://www.fujitsu.com/global/digitalannealer/pdf/wp-da-financialsector-ww-en.pdf
“The Diamond Quantum Revolution” Physics World. 2020 [WEB POST]. https://physicsworld.com/a/the-diamond-quantum%E2%80%AFrevolution/
“The Next Decade in Quantum Computing—and How to Play” by BCG. 2018 [REPORT]. https://www.bcg.com/publications/2018/next-decade-quantum-computing-how-play
“The Pharmaceutical Industry in Figures” European Federation of Pharmaceutical Industries and Associations. 2023 [WEB POST]. https://www.efpia.eu/media/637143/the-pharmaceutical-industry-in-figures-2022.pdf
“The Physical Implementation of Quantum Computation” by David P. DiVincenzo. arXiv. 2000 [RESEARCH PAPER]. https://arxiv.org/abs/quant-ph/0002077
“The Quantum Revolution: Networking and Security for Tomorrow’s Internet” by CISCO. 2023 [WEB POST]. https://newsroom.cisco.com/c/r/newsroom/en/us/a/y2023/m06/the-quantum-revolution-networking-and-security-for-tomorrows-internet.html
“The Quantum Tortoise and the Classical Hare: A Simple Framework for Understanding Which Problems Quantum Computing Will Accelerate (And Which It Will Not)” by Sukwoong Choi, William S. Moses and Neil Thompson. arXiv. 2023 [RESEARCH PAPER]. https://arxiv.org/abs/2310.15505
“The Quest for the Ultimate Seed Begins With Quantum Computing” Global AgInvesting. 2021 [WEB POST]. https://www.globalaginvesting.com/contributed-content-quest-ultimate-seed-begins-quantum-computing/
“The Race toward Quantum Advantage” Semiconductor Engineering. 2023 [WEB POST]. https://semiengineering.com/the-race-toward-quantum-advantage/
“The Second Quantum Revolution” NIST. 2018 [WEB POST]. https://www.nist.gov/physics/introduction-new-quantum-revolution/second-quantum-revolution
“Thinking through the Ethics of New Tech…Before There’s a Problem” by Beena Ammanath. Harvard Business Review Publishing. 2021 [WEB ARTICLE]. https://hbr.org/2021/11/thinking-through-the-ethics-of-new-techbefore-theres-a-problem
“Total is Exploring Quantum Algorithms to Improve CO2 Capture” Total Energies. 2020 [WEB POST]. https://totalenergies.com/media/news/news/total-exploring-quantum-algorithms-improve-co2-capture
“Towards the Industrialisation of Quantum Key Distribution in Communication Networks: A Short Survey” by Ruiqi Liu et al. Wiley. 2022 [RESEARCH PAPER].
“Toyota Central R&D Labs: A Quantum Approach to Transportation” D-Wave. 2021 [CASE STUDY]. https://www.dwavesys.com/media/b4lllchm/dwave_toyota_case_story_v4.pdf
“Toyota Central R&D Labs: A Quantum Approach to Transportation” DWave. 2021 [CASE STUDY]. https://www.dwavesys.com/media/b4lllchm/dwave_toyota_case_story_v4.pdf
“Toyota Harnesses Quantum Computers to Develop battery Materials” Asia Nikkei. 2021 [PRESS RELEASE]. https://asia.nikkei.com/Business/Technology/Toyota-harnesses-quantum-computers-to-develop-battery-materials
“Toyota Quantum Annealing of Vehicle Routing Problem with Time, State and Capacity” by Hirotaka Irie et al. arXiv. 2019 [RESEARCH PAPER]. https://arxiv.org/pdf/1903.06322.pdf
“US Defense Spending Compared to Other Countries” Peter G. Foundation. 2023 [WEB POST]. https://www.pgpf.org/chart-archive/0053_defense-comparison
“US Navy, Air Force Collaborate on Quantum” IoT World Day. 2023 [WEB POST]. https://www.quantumbusinessnews.com/applications/us-navy-air-force-collaborate-on-quantum

“Using Quantum Computing to Tell the Weather” Azo Quantum. 2018 [WEB POST]. https://www.azoquantum.com/Article.aspx?ArticleID=98#:~:text=Quantum%20Computing%20Improves%20Weather%20Forecasting&text=It%20employs%20IBM’s%20supercomputing%20technology,supercomputers%20are%20unable%20to%20achieve

“Using Zapata’s Quantum Workflows Platform, Orquestra, KAUST Will Explore How Quantum Computing Can Simulate and Optimize the Aerodynamic Design Process for Vehicles” Zapata. [WEB POST]. https://zapata.ai/news/zapata-partner-with-kaust-to-bring-quantum-computing-to-the-middle-east-for-the-advancement-of-computational-fluid-dynamics/

“Variational Quantum Algorithms” by M. Cerezo et al. Nature. 2021 [RESEARCH PAPER]. https://www.nature.com/articles/s42254-021-00348-9, https://arxiv.org/pdf/2012.09265.pdf

“Volkswagen Demonstrates First Successful Real-world Use of Quantum Computing to Help Optimize Traffic Routing” Green Car Congress. 2019 [WEB POST]. https://www.greencarcongress.com/2019/12/20191206-vwquantum.html

“Volkswagen Has Developed a Quantum-inspired Algorithm for Automotive Image Recognition That Is More Effective Than Classical Machine Learning Neural Networks” Eenews. 2023 [WEB POST]. https://www.eenewseurope.com/en/quantum-algorithm-boost-for-automotive-image-sensors/

“Volkswagen: Navigating Tough Automotive Tasks with Quantum Computing” Dwave. 2021 [CASE STUDY]. https://www.dwavesys.com/media/2pojgtcx/dwave_vw_case_story_v2f.pdf

“What Does the Future of Worker-Management Dialogue Look Like?” The Threads. [WEB POST]. https://thethreads.org/contribution/what-does-the-future-of-worker-management-dialogue-look-like/

“What Is an Economic Sector and How Do the 4 Main Types Work?”: https://www.investopedia.com/terms/s/sector.asp

“What Is Cryptographic Agility? How to Get Crypto-Agility?” Encryption Consulting. [WEB POST]. https://www.encryptionconsulting.com/education-center/what-is-crypto-agility

“What Is Hilbert Space?” [WEB POST]. https://www.scienceabc.com/pure-sciences/what-is-a-hilbert-space.html

“What Is Quantum Annealing?” D-WAVE. [WEB POST]. https://docs.dwavesys.com/docs/latest/c_gs_2.html

“What Is Quantum Processing Unit (QPU)” Huawei. 2023 [WEB POST]. https://forum.huawei.com/enterprise/en/what-is-quantum-processing-unit-qpu/thread/667282964481982464-667213859733254144

“What Is Quantum Sensing?” Bae Systems. [WEB POST]. https://www.baesystems.com/en-us/definition/what-is-quantum-sensing

“What Is Quantum-safe Cryptography?” IBM. [WEB POST]. https://www.ibm.com/topics/quantum-safe-cryptography

“WSJ: Accenture, 1QBit and Biogen POC Shows Quantum Computing May Speed Drug Discovery” 1Qbit. 2017 [PRESS RELEASE]. https://1qbit.com/news/wsj-accenture-1qbit-biogen-poc-indicates-quantum-computing-may-speed-drug-discovery/

“Xanadu and Rolls-Royce to Build Quantum Computing Tools with PennyLane” Yahoo! Finance. 2023 [NEWS POST]. https://finance.yahoo.com/news/xanadu-rolls-royce-build-quantum-130000758.html